

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279341

Kind Code

A1

Publication Date

September 04, 2025

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Multilayer Cores, Variable Width Vias, and Offset Vias

Abstract

Novel tools and techniques are provided for implementing a semiconductor package or a chip package, and more particularly, for implementing a semiconductor package or a chip package including a core or a multilayer core having one or more variable width vias or one or more offset vias. In various embodiments, an apparatus includes a substrate. The substrate includes a core. The core may include one or more vias extending through the core. At least one via of the one or more vias includes a cross-section that varies along a length of the at least one via as the via extends through the core. The cross-section of the via may vary based on at least one of varying a width of the at least one via or offsetting a first portion of the at least one via from a second portion of the at least one via.

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Family ID: 88414371

Appl. No.: 19/213327

Filed: May 20, 2025

Related U.S. Application Data

parent US division 17732954 20220429 PENDING child US 19213327

Publication Classification

Int. Cl.: H01L23/498 (20060101); H01L21/48 (20060101); H05K1/02 (20060101); H05K1/11 (20060101); H05K3/00 (20060101); H05K3/42 (20060101)

CPC **H01L23/49822** (20130101); **H01L21/4857** (20130101); **H01L23/49827** (20130101);
H01L23/49838 (20130101); **H05K1/0298** (20130101); **H05K1/115** (20130101);
H05K3/0047 (20130101); **H05K3/429** (20130101);

Background/Summary

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FIELD

[0002] The present disclosure relates, in general, to methods, systems, and apparatuses for implementing a semiconductor package or a chip package, and more particularly to methods, systems, and apparatuses for implementing a semiconductor package or a chip package including a core or a multilayer core having one or more variable width vias or one or more offset vias.

BACKGROUND

[0003] Chip packages are ever increasing in size requiring thicker cores (e.g., the dielectric material or dielectric layer of a substrate) to control package warpage. However, having thicker cores in chip packages results in longer core vias (e.g., plated through holes (“PTHs”)) extending through the cores of the chip packages. There are several disadvantages with having longer core vias. Longer core vias cause larger impedance variations as electrical signals travel through the chip package. This in turn results in higher return loss and eventual degradation of signal integrity. In addition, thicker cores with longer core vias cause resonance in insertion loss plots at frequencies which fall in the bandwidth of interest of todays high-speed Serializer/Deserializers (“SerDes”) and cannot be rectified with available organic structure technology.

[0004] Hence, there is a need for more robust and scalable solutions for implementing semiconductor packages and chip packages. Thus, methods, systems, and apparatuses for implementing semiconductor packages or chip packages including a core or a multilayer core having one or more variable width vias or one or more offset vias are provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] A further understanding of the nature and advantages of particular embodiments may be realized by reference to the remaining portions of the specification and the drawings, in which like reference numerals are used to refer to similar components. In some instances, a sub-label is associated with a reference numeral to denote one of multiple similar components. When reference is made to a reference numeral without specification to an existing sub-label, it is intended to refer to all such multiple similar components.

[0006] FIGS. 1A and 1B are schematic cross-sectional views of a substrate of a semiconductor or chip module comprising a multilayer core, in accordance with various embodiments;

[0007] FIG. 1C is a top view of a substrate of a semiconductor or chip module, in accordance with various embodiments;

[0008] FIGS. 2A-2D are schematic cross-sectional views of a substrate of a semiconductor or chip module comprising a variable width via, in accordance with various embodiments;

[0009] FIGS. 3A and 3B are schematic cross-sectional views of a substrate of a semiconductor or chip module comprising a multilayer core layer and a variable width via, in accordance with various embodiments;

[0010] FIGS. 4A and 4B are schematic cross-sectional views of a substrate of a semiconductor or chip module comprising at least two variable width vias, in accordance with various embodiments;

[0011] FIGS. 5A and 5B are schematic cross-sectional views of a substrate of a semiconductor or chip module comprising an offset via, in accordance with various embodiments;

[0012] FIGS. 6A and 6B are schematic cross-sectional views of a substrate of a semiconductor or chip module comprising a multilayer core and an offset diameter via, in accordance with various embodiments;

[0013] FIGS. 7A and 7B are schematic cross-sectional views of a substrate of a semiconductor or chip module comprising at least two offset vias, in accordance with various embodiments; and

[0014] FIG. 8 is a flow diagram of a method of manufacturing a substrate of a semiconductor or chip module comprising a variable width via or an offset via, in accordance with various embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

[0015] Various embodiments provide tools and techniques for implementing semiconductor packages or chip packages including a multilayer core and/or a core having one or more variable width vias or one or more offset vias.

[0016] In a first aspect, an apparatus including a substrate is provided. The substrate might be a substrate of a printed circuit board (“PCB”), a substrate of an integrated circuit package (e.g., ball grid array (BGA) package, a pin grid array (PGA) package, or other integrated circuit package type that includes a substrate), and/or any other substrate upon which elements of a semiconductor or chip module are fabricated or attached. The substrate includes a core comprising one or more vias extending through the core. The one or more vias may extend along an axis normal to a plane defined by a planar surface of the core. At least one via of the one or more vias has a cross-section that varies along a length of the at least one via as the at least one via extends through the core. In some instances, the cross-section of the at least one via varies based on at least one of varying a width of the at least one via or offsetting a first portion of the at least one via from a second portion of the at least one via.

[0017] In some embodiments, varying the width of the at least one via comprises at least one of a middle portion of the at least one via of the one or more vias having a greater width than at least one of a top portion or a bottom portion of the at least one via. At least one of the top portion and the bottom portion of the at least one via might have a same width. Alternatively, at least one of the top portion and the bottom portion of the at least one via have a different width.

[0018] In some instances, the one or more vias comprise at least one pair of vias. A first cross-section of a first via of the pair of vias varies along a first length of the first via based on a varying a first width of the first via and a second cross-section of a second via of the pair of vias varies along a second length of the second via based on varying a second width of the second via. In some instances, the first width and the second width may be a same width. Alternatively, the first width and the second width may be a different width. In various embodiments, a first middle portion of the first via of the pair of vias has a greater width than at least one of a first top portion or a first bottom portion of first via and a second middle portion of the second via of the pair of vias has a greater width than at least one of a second top portion or a second bottom portion of the second via.

[0019] In various embodiments, the one or more vias comprise at least one pair of vias. A first cross-section of a first via of the pair of vias varies along a first length of a first via of the pair of vias based on offsetting a first portion of the first via from a second portion of the first via and a second cross-section of a second via of the pair of vias varies along a second length of a second via of the pair of vias based on offsetting a first portion of the second via from a second portion of the second via. The second portion of the first via is offset toward the second portion of the second via

and the second portion of the second via is offset toward the second portion of the first via. In some instances, the second portion of the first via is a first middle portion of the first via and the second portion of the second via is a second middle portion of the second via and the first middle portion of the first via is offset toward the second middle portion of the second via and the second middle portion of the second via is offset toward the first middle portion of the first via. The first middle portion of the first via may be a same length as the second middle portion of the second via.

[0020] In another aspect, a semiconductor device is provided. The semiconductor device includes a substrate comprising two or more core layers. Each core layer is stacked on top of or below another core layer of the two or more core layers. Each core layer may be stacked along an axis normal to a plane defined by a planar surface of each of the two or more core layers. Each core layer includes at least one drill hole. At least one first drill hole of at least one first core layer electrically connects with at least one other drill hole of at least one other core layer to form a via through the two or more core layers. The via has a cross-section that varies along a length of the via as the via extends through the two or more core layers. The cross-section of the via varies based on at least one of having a first width of at least one first drill hole different from a second width of the at least one other drill hole or offsetting a first portion of the at least one first drill hole from a second portion of the at least other drill hole.

[0021] In various cases, the at least one first core layer of the two or more core layers comprises a different thickness from the at least one other core layer. The at least one first core layer of the two or more core layers may be formed from a material with a different dielectric coefficient than the at least one other core layer. In some examples, the two or more core layers comprise three core layers and a thickness of each core layer of the three core layers is in the range of 385 micrometers to 435 micrometers.

[0022] In some embodiments, the two or more core layers comprise one or more top core layers, one or more middle core layers, and one or more bottom core layers. Varying the cross-section along the length of the via comprises having a larger middle width of one or more first middle drill holes of the one or more middle core layers than a bottom width of one or more bottom drill holes of the one or more bottom core layers or a top width of one or more top drill holes of the one or more top core layers. The one or more middle drill holes, the one or more bottom drill holes, and the one or more top drill holes electrically connect together to form the via extending through the two or more core layers. In some cases, the bottom width of the one or more bottom drill holes of the one or more bottom core layers is the same as the top width of the one or more top drill holes of the one or more top core layers. Alternatively, in other cases, the bottom width of the one or more bottom drill holes of the one or more bottom core layers is different than the top width of the one or more top drill holes of the one or more top core layers.

[0023] In various instances, each core layer comprises a pair of drill holes. A first drill hole of a first pair of drill holes of the at least one first core layer electrically connects with a first other drill hole of a second pair of drill holes of the at least one other core layer to form a first via through the two or more core layers. A first cross-section of the first via varies based on at least one of having a first width of the first drill hole different from a first other width of the first other drill hole or offsetting the first drill hole from the first other drill hole. A second drill hole of the first pair of drill holes of the at least one first core layer electrically connects with a second other drill hole of the second pair of drill holes of the at least one other core layer to form a second via through the two or more core layers. A second cross-section of the second via varies based on at least one of having a second width of the second drill hole different from a second other width of the second other drill hole or offsetting the second drill hole from the second other drill hole. When the first drill hole is offset from the first other drill hole and the second drill hole is offset from the second other drill hole, the first drill hole is offset toward the second drill hole and the second drill hole is offset toward the first drill hole. When the first drill hole is offset from the first other drill hole and the second drill hole is offset from the second other drill hole, the first drill hole is electrically

connected to the first other drill hole using a first conductive metal running along a first parallel axis parallel to the plane defined by the planar surface of each of the two or more core layers, and wherein the second drill hole is electrically connected to the second other drill hole using a second conductive metal running along a second parallel axis parallel to the plane defined by the planar surface of each of the two or more core layers.

[0024] In yet another aspect, a method of manufacturing a variable width via or offset via is provided. The method includes forming at least one first core layer of a core comprising at least one first drill hole. The method continues with forming at least one other core layer of the core comprising at least one other drill hole. The at least one other core layer is layered on top of or below the at least one first core layer along an axis normal to a plane defined by a planar surface of the at least one first core layer. The method further includes electrically coupling the at least one first drill hole with the at least one other drill hole to create a via through the core. The via has a cross-section that varies along a length of the via as the via extends through the core. The cross-section of the via varies based on at least one of having a first width of at least one first drill hole different from a second width of the at least one other drill hole or offsetting a first portion of the at least one first drill hole from a second portion of the at least other drill hole.

[0025] In the following description, for the purposes of explanation, numerous details are set forth to provide a thorough understanding of the described embodiments. It will be apparent to one skilled in the art, however, that other embodiments may be practiced without some of these details. Several embodiments are described herein, and while various features are ascribed to different embodiments, it should be appreciated that the features described with respect to one embodiment may be incorporated with other embodiments as well. By the same token, however, no single feature or features of any described embodiment should be considered essential to every embodiment of the invention, as other embodiments of the invention may omit such features.

[0026] Similarly, when an element is referred to herein as being “connected” or “coupled” to another element, it is to be understood that the elements can be directly connected to the other element, or have intervening elements present between the elements. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, it should be understood that no intervening elements are present in the “direct” connection between the elements. However, the existence of a direct connection does not exclude other connections, in which intervening elements may be present.

[0027] Furthermore, the methods and processes described herein may be described in a particular order for ease of description. However, it should be understood that, unless the context dictates otherwise, intervening processes may take place before and/or after any portion of the described process, and further various procedures may be reordered, added, and/or omitted in accordance with various embodiments.

[0028] Unless otherwise indicated, all numbers used herein to express quantities, dimensions, and so forth should be understood as being modified in all instances by the term “about.” In this application, the use of the singular includes the plural unless specifically stated otherwise, and use of the terms “and” and “or” means “and/or” unless otherwise indicated. Moreover, the use of the term “including,” as well as other forms, such as “includes” and “included,” should be considered non-exclusive. Also, terms such as “element” or “component” encompass both elements and components comprising one unit and elements and components that comprise more than one unit, unless specifically stated otherwise.

[0029] Additionally, it should be understood that spatial descriptions (e.g., “above,” “below,” “up,” “left,” “right,” “down,” “top,” “bottom,” “middle,” “vertical,” “horizontal,” etc.) used herein are for purposes of illustration only, and that practical implementations of the structures described herein can be spatially arranged in any orientation or manner.

[0030] In conventional semiconductor or chip packages, one or more vias may extend through a single layer core of the semiconductor or chip package. The one or more vias (e.g., a plated through

hole (“PTH”) via) are electrically conductive openings through a core of the semiconductor or chip package through which an electrical signal may pass. The one or more vias pass one or more electrical signals between different parts of the semiconductor or chip package separated by the single layer core. Conventionally, the one or more vias that extend through the single layer core of the semiconductor or chip package have a same width or a same diameter centered around a vertical axis extending through the center of the via.

[0031] The proposed module may be a semiconductor or chip module (e.g., integrated circuit (IC), chip, or other semiconductor device) that provides a multilayer core (e.g., multiple dielectric layers of a substrate), that provides a variable width or a variable diameter via, and/or that provides a first portion of the via that is offset from a second portion of the via. By implementing multilayer cores, variable width vias, and/or offset vias, impedance variations seen by the electrical signal as it travels through the semiconductor or chip package may be minimized. This in turn may improve insertion and return loss of the semiconductor or chip package. Further, by implementing multilayer cores, variable width vias, and/or offset vias, resonance seen in the insertion loss in the frequency bandwidth of interest and cross talk may be minimized or eliminated. Additionally, by implementing a multilayer core, package warpage may be reduced.

[0032] FIGS. 1A and 1B (collectively, FIG. 1) are schematic cross-sectional side views of a substrate **100a** and **100b** (collectively, substrate **100**) of a semiconductor or chip module, in accordance with various embodiments. FIG. 1C is a top view of the substrate **100**, in accordance with various embodiments. The semiconductor modules may be an integrated circuit (IC), chip, or other semiconductor device, and/or the like. The substrate **100** may be a substrate of a printed circuit board (“PCB”), a substrate of an integrated circuit package (e.g., ball grid array (BGA) package, a pin grid array (PGA) package, or other integrated circuit package type that includes a substrate), and/or any other substrate upon which elements of a semiconductor or chip module are fabricated or attached. The substrate **100** may have various other components (e.g., components represented in FIGS. 2-7 and/or other components known to persons of ordinary skill in the art) than those depicted in FIG. 1 and is not limited to only the components depicted in FIG. 1.

[0033] The substrate **100** may include a first layer **105** and a second layer **110**. The first layer **105** and the second layer **110** might be one or more surface layers of the substrate **100**. Alternatively, the first layer **105** and the second layer **110** might be one or more inner layers of the substrate **100**. The first layer **105** and/or the second layer **110** might have one or more conductive components (e.g., pins, pads, solder balls, etc.) located on the first layer **105** or the second layer **110**, may be coated in a conductive coating, may be formed from a conductive material, and/or the like.

[0034] A core **115** is located between the first layer **105** and the second layer **110** of the substrate **100**. The core **115** of the substrate has one or more dielectric layers **120a**, **120b**, and **120c** (collectively, core layers **120**). Although three core layers **120** are shown in FIG. 1, there may be more or less core layers **120** located in substrate **100**.

[0035] In some cases, each core layer **120a**, **120b**, and **120c** may be formed from a same dielectric material or, in other cases, at least one core layer **120a**, **120b**, or **120c** may be formed from a different dielectric material. In various embodiments, each core layer **120a**, **120b**, and **120c** may be formed from a material with a same dielectric coefficient or, in other cases, at least one core layer **120a**, **120b**, or **120c** may be formed from a material with a different dielectric coefficient.

[0036] In some cases, as shown in FIG. 1B, each core layer **120a**, **120b**, and **120c** may be a same or similar thickness. Alternatively, a bottom core layer **120a** and a top core layer **120c** may be a same or a similar thickness and a middle core layer **120b** may be larger (as shown in FIG. 1A) or smaller than the bottom core layer **120a** and the top core layer **120c**. Alternatively, at least one core layer **120a**, **120b**, or **120c** may be a different thickness from the other core layers. In one embodiment, the thickness of each core layer **120a**, **120b**, and **120c** may be in the range of 385 micrometers to 435 micrometers.

[0037] Each core layer **120a**, **120b**, and **120c** may have a corresponding drill hole **125a**, **125b**, and

125c. Although three drill holes **125a**, **125b**, and **125c** are shown in FIG. 1, there may be more or less drill holes located in each core layer **120a**, **120b**, and **120c**. Each drill hole **125a-125c** may have a corresponding length. For example, as shown in FIG. 1A, drill hole **125a** may have a length **L1**, drill hole **125b** might have a length **L2**, and drill hole **125c** might have a length **L3**. Although lengths **L1-L3** are shown in FIG. 1A, a person of ordinary skill in the art would know how to identify the lengths of the drill holes in FIG. 1B and the remaining figures (e.g., FIGS. 3, 4B, 6, and 7B). A length of each drill hole **125a**, **125b**, and **125c** may vary based on the thickness or a length of each core layer **120a-120c**. For example, if a middle core layer **120b** is thicker than layers **120a** and **120c**, then the length **L2** of middle drill hole **125b** might be longer than the length **L1** of bottom drill hole **125a** and the length **L3** of top drill hole **125c**.

[0038] The three drill holes **125a**, **125b**, and **125c** might electrically couple together to form a via **130** through the core layers **120a**, **120b**, and **120c** of the substrate **100**. The via **130** may extend along an axis A-A (shown in FIGS. 1A and 1B) which is normal to a plane defined by a planar surface of the core **115**. The via **130** may extend completely through the substrate **100** (e.g., a “through” via), may extend through a portion of the substrate **100** from one outer surface (e.g., a “blind” via), or may extend through a portion of the substrate **100** and be completely hidden from external view (e.g., a “buried” via), and/or the like. It should be noted that the core **115** is not limited to only one via **130** and may include more than one via as shown in FIGS. 4 and 7. Additionally, it should also be noted that the various components of the substrate **100**, core **115**, and via **130** are schematically illustrated in FIG. 1, and that modifications to the various components and other arrangements of the substrate **100**, core **120**, and via **130** may be possible and in accordance with the various embodiments.

[0039] The drill holes **125a**, **125b**, and **125c** may be formed in each core layer **120a**, **120b**, and **120c** by drilling a hole in each core layer **120a**, **120b**, and **120c**, and plating or coating an inner surface of the hole with an electrically conductive material. The drill holes **125a**, **125b**, and **125c** may be plated or coated with a metal such as copper, aluminum, gold, silver, tin, nickel, lead, or a combination of metals/alloy, or may be formed of other electrically conductive material.

[0040] The via **130** may be formed by forming a first core layer **120a** and drilling a first drill hole **125a** in the first core layer **120a**. Next, a second core layer **120b** may be formed on top of the first core layer **120a** and a second drill hole **125b** may be drilled in the second core layer **120b** and electrically coupled to the first drill hole **125a** of the first core layer **120a**. Additionally, a third core layer **120c** may be formed on top of the second core layer **120b** and a third drill hole **125c** may be drilled in the third core layer **120c** and electrically coupled to the second drill hole **125b** of the second core layer **120b**. The drill holes **125a-125c** may be electrically coupled together using an electrically conductive material. The electrically conductive material plating or coating the drill holes **125a**, **125b**, and **125c** forms the via **130** which creates an electrical connection for an electrical signal to travel from the first layer **105** to the second layer **110** of the substrate **100**. In some cases, there may be other ways or additional methods to form via **130** and the formation of via **130** is not limited to any particular method of manufacturing a particular core.

[0041] In some instances, as shown in FIG. 1B, the first drill hole **125a** might optionally include a first via pad **135a** and a second via pad **135b**, the second drill hole **125b** might optionally include the second via pad **135b** and a third via pad **135c**, and the third drill hole **125c** might optionally include the third via pad **135c** and a fourth via pad **135d**. First via pad **135a** is formed on a first layer **105** of the substrate **100** around or near an opening of the first drill hole **125a**, second via pad **135b** is formed between the first drill hole **125a** and the second drill hole **125b**, third via pad **135c** is formed between the second drill hole **125b** and the third drill hole **125c**, and the fourth via pad **135d** is formed on the second layer **110** of the substrate **100** around or near the opening of the third drill hole **125c**.

[0042] Via pads **135a-135d** are electrically coupled to drill holes **125a-125c** and/or via **130**. In some instances, vias pads **135a-135d** surround an opening of drill holes **125a-125c** to electrically

couple drill holes **125a-125c** together and form via **130**. Additionally or alternatively, via pads **135a-135d** run along a first parallel axis arranged parallel to the plane defined by the planar surface on top of a corresponding core layer **120a-120c** or between two corresponding core layers to electrically couple drill holes **125a-125c** together and form via **130**. In some examples, via pads **135a-135d** may be formed of a metal such as copper, aluminum, gold, silver, tin, nickel, lead, or a combination of metals/alloy, or may be formed of other electrically conductive material.

[0043] In some instances, as shown in FIG. 2C for drill hole **125c**, via **130**, and via pad **135c**, the openings of drill holes **125a**, **125b**, and **125c**, the opening of via **130**, and/or via pads **135a-135d** might be circle-shaped, oval-shaped, square-shaped, rectangular-shaped, and/or the like. Typically, the openings of drill holes **125a**, **125b**, and **125c** and/or the opening of via **130** will be circle-shaped or oval-shaped. The via **130** includes a first portion or bottom portion (e.g., bottom drill hole **125a**), a second portion or middle portion (e.g., middle drill hole **125b**), and a third portion or top portion (e.g., top drill hole **125c**). In some cases, as shown in FIGS. 1A and 1B, a cross-section of the via **130** is constant throughout the three drill holes **125a**, **125b**, and **125c** as the via **130** extends through core **115**.

[0044] By creating a via **130** using a multilayer core **115**, warpage of a semiconductor package and/or chip package may be reduced because a multilayer core **115** provides more stability to substrate **100**. Further, by creating each drill hole **125a**, **125b**, and **125c** layer by layer manufacturing processes can use a smaller drill and create smaller cuts in each core layer **120a**, **120b**, and **120c**. This in turn reduces cross-talk because there can be more space between the traces. Additionally, having thinner core layers **120a**, **120b**, and **120c** instead of a thicker one layer core, helps reduce resonance of the core which improves insertion loss and return loss.

[0045] Different arrangements of substrate **100**, core **115**, and via **130** are illustrated below with respect to FIGS. 2-7.

[0046] FIGS. 2A-2D (collectively, FIG. 2) are schematic cross-sectional views of a substrate **200a-200d** (collectively, substrate **200**) of a semiconductor or chip module, in accordance with various embodiments. The substrate **200** may be similar to the substrate **100** of FIG. 1 with a few additional and/or different features. In some cases, functionalities, features, modifications, alterations and/or the like described in FIG. 1 may also be applied to FIG. 2 and vice versa. It should be noted that the various components of the substrate **200** are schematically illustrated in FIG. 2, and that modifications to the various components and other arrangements of the substrate **200** may be possible and in accordance with the various embodiments.

[0047] The substrate **200** may include a first layer **205** (e.g., similar to first layer **105** of FIG. 1) and a second layer **210** (e.g., similar to second layer **110** of FIG. 1). A core **215** is located between the first layer **205** and the second layer **210** of the substrate **200**. The core **215** may be formed from one or more dielectric materials and/or dielectric layers. In FIG. 2, the core **215** is only one layer, but is not limited to only one layer, as shown in FIGS. 1, 3, 4, 6, and 7.

[0048] The core **215** may include a via **220** extending through the core **215**. The via **220** may extend along an axis A-A (shown in FIG. 2A) which is normal to a plane defined by a planar surface of the core **215**. It should be noted that the core **215** is not limited to only one via **220** and may include more than one via **220** as shown in FIGS. 4 and 7.

[0049] The via **220** may be formed in the core **215** by drilling a hole in the core **215** and plating or coating an inner surface of the hole with an electrically conductive material. The electrically conductive material may be copper, aluminum, gold, silver, tin, nickel, lead, or a combination of metals/alloy, or may be formed of other electrically conductive material. The electrically conductive material plating or coating the inner surface of the via **220** forms an electrical connection from the first layer **205** to the second layer **210** of the substrate **200**.

[0050] In some instances, the opening or drill hole **225** of via **220** might be circle-shaped, oval-shaped, square-shaped, rectangular-shaped, and/or the like. Typically, the opening or drill hole **225** of the via **220** will be circle-shaped or oval-shaped. The via **220** includes a first portion or bottom

portion **230**, a second portion or middle portion **235**, and a third portion or top portion **240**. In some cases, the first portion **230** has a first width or a first diameter **W1** (shown in FIG. 2B), the second portion **235** has a second width or a second diameter **W2** (shown in FIG. 2B), and the third portion **240** has a third width or a third diameter **W3** (shown in FIG. 2B). Although the widths (**W1-W3**) are only shown in FIG. 2B, a person of ordinary skill in the art would know how to identify the widths in FIGS. 2A, 2B, and 2C and the remaining figures (e.g., FIGS. 1 and 3-7). The via **220** is not limited to only three portions and/or three widths and may have more or less than three portions and/or three widths.

[0051] In some embodiments, a cross-section of the via **220** might vary along the length **Ly** of the via **220** as the via **220** extends through the core **215**. The cross-section of the via **220** may vary due to varying or variable widths of the via **220**. In some cases, the first width (or bottom width) **W1** of the bottom portion **230** of the via **220** and the third width (or top width) **W3** or the top portion **240** of the via **220** might be a same width or approximately the same width. Alternatively, in other cases, the first width (or bottom width) **W1** of the bottom portion of the via **220** and the third width (or top width) **W3** of the top portion **240** of the via **220** might be a different width. The second width (or middle width) **W2** of the middle portion **235** of the via **220** is different from the first width (or bottom width) **W1** of the bottom portion **230** of the via **220** and the third width (or top width) **W3** of the top portion **240** of the via **220**. The second width (or middle width) **W2** of the middle portion **235** of the via **220** is larger or has a greater width than the first width (or top width) **W1** of the bottom portion **230** of the via **220** and the third width (or bottom width) **W3** of the top portion **240** of the via **220**. In various instances, as shown in FIGS. 2C and 2D, the width of the via **220** may vary along a slope (e.g., slopes **245**, **250**, **255**, and/or **260**) or a curve (e.g., curves **265** and/or **270**) as the length **Ly** of the via **220** extends through the core **215**. In some cases, the width of the via **220** may gradually increase as the via **220** approaches a middle or approximately a middle of the substrate **200**, the core **215**, and/or the via **220** and gradually decrease as the via **220** approaches an end of the substrate **200**, the core **215**, and/or the via **220**, such that a middle portion **235** of the via **220** is larger than a bottom portion **230** of the via **220** and/or a top portion **240** of the via **220**.

[0052] In some cases, as shown in FIG. 2A, a length of each of the first portion **230**, second portion **235**, and third portion **240** of via **220** may be the same or approximately the same. Alternatively, as shown in FIG. 2B, a length of each of the first portion **230**, second portion **235**, and third portion **240** may vary. For example, as shown in FIG. 2B, the length of the middle portion **235** of the via **220** is shown as being longer than the length of the top portion **240** of the via **220** and the bottom portion **230** of the via **220**. Alternatively, in some cases, the length of the middle portion **235** of the via **220** may be shorter than the length of the top portion **240** of the via **220** and the bottom portion **230** of the via **220**. Each of the lengths of the first portion **230**, the second portion **235**, and the third portion **240** of the via **220** may be adjusted or varied to reduce the impedance variation of the substrate **200**. In some cases, when core **215** is a multilayer core, the length of the portions of via **220** might vary based on the thickness of each core layer.

[0053] It should be noted that the via **220** is not limited to only the shapes, widths, lengths, modifications, or arrangements shown in FIGS. 2A-2D, but may also be various other shapes, widths, lengths, modifications, or arrangements with variable widths, in accordance with the various embodiments. In addition, a person of ordinary skill in the art would understand that the shapes, widths, lengths, modifications, or arrangements shown in FIGS. 2A-2D may also be applied to the substrates shown in FIGS. 1 and 3-7. However, FIGS. 1 and 3-7 are also not limited to the various other shapes, widths, lengths, modifications, or arrangements shown, but may also be various other shapes, widths, lengths, modifications, or arrangements. Each of the shapes, widths, lengths, of the vias shown in FIGS. 1 and 3-7 may be adjusted or arranged to reduce the impedance variation of the substrate substrates or vias.

[0054] By varying the width of the via **220** such that the via **225** has a larger width in a middle

portion **235** of the via **220**, impedance variations seen by the electrical signal as it travels through the substrate **200** by the via **220** may be minimized. For example, by having a bigger diameter in the middle portion **235** of the via **220** and smaller diameter at the two ends (e.g., bottom portion **230** and top portion **240**) of the via **220**, this arrangement helps to minimize the inductive peak seen in the impedance variation profile of the substrate **200**. This in turn may improve insertion and return loss of the substrate **200**.

[0055] Different arrangements of substrate **200**, core **215**, and via **220** are illustrated below with respect to FIGS. 3-7.

[0056] FIGS. 3A and 3B (collectively, FIG. 3) are schematic cross-sectional views of a substrate **300a** and **300b** (collectively, substrate **300**) of a semiconductor or chip module, in accordance with various embodiments. The substrate **300** may be similar to the substrate **100** of FIG. 1 and/or the substrate **200** of FIG. 2 with a few additional and/or different features. In some cases, functionalities, features, modifications, alterations and/or the like described in FIGS. 1 and 2 may also be applied to FIG. 3 and vice versa. It should be noted that the various components of the substrate **300** are schematically illustrated in FIG. 3, and that modifications to the various components and other arrangements of the substrate **300** may be possible and in accordance with the various embodiments.

[0057] The substrate **300** may include a first layer **305** (similar to first layer **105** of FIG. 1 and **205** of FIG. 2) and a second layer **310** (similar to second layer **110** of FIG. 1 and **210** of FIG. 2). A core **315** is located between the first layer **305** and the second layer **310** of the substrate **300**. The core **315** may have one or more layers **320a**, **320b**, and **320c** (collectively, core layers **320**) similar to the core layers **120a**, **120b**, and **120c** of FIG. 1. Although three core layers **320** are shown in FIG. 3, there may be more or less core layers **320** located in substrate **300**. By having a multilayer core **315**, cross talk and return loss of the substrate **300** may be minimized or reduced. In addition, having a multilayer core **315** may increase the rigidity of the substrate **300** and reduce package warpage.

[0058] Each core layer **320a**, **320b**, and **320c** may have a corresponding drill hole **325a**, **325b**, and **325c**. Although three drill holes **325a**, **325b**, and **325c** are shown in FIG. 3, there may be more or less drill holes located in each core layer **320a**, **320b**, and **320c**. The three drill holes **325a**, **325b**, and **325c** might electrically couple together to form a via **330** through the core layers **320a**, **320b**, and **320c**. The via **330** may extend along an axis A-A (shown in FIG. 3A) which is normal to a plane defined by a planar surface of the core **315**. It should be noted that the core **315** is not limited to only one via **330** and may include more than one via. Each core layer **320**, drill hole **325a-325c**, and via **330** may be formed in a manner similar to the manufacturing method recited in FIG. 8.

[0059] In some instances, the first drill hole **325a** might optionally include a first via pad **335a** and a second via pad **335b**, the second drill hole **325b** might optionally include the second via pad **335b** and a third via pad **335c**, and the third drill hole **325c** might optionally include the third via pad **335c** and a fourth via pad **335d**. Via pads **335a-335d** are electrically coupled to drill holes **325a-325c** and/or via **330**. In some instances, vias pads **335a-335d** run along a first parallel axis arranged parallel to the plane defined by the planar surface on top of a corresponding core layer **320a-320c** to electrically couple drill holes **325a-325c** together and form via **330**.

[0060] In some instances, the openings of drill holes **325a**, **325b**, and **325c**, the opening of via **330**, and/or via pads **335a-335d** might be circle-shaped, oval-shaped, square-shaped, rectangular-shaped, and/or the like. The via **330** includes a first portion or bottom portion (e.g., bottom drill hole **325a**), a second portion or middle portion (e.g., middle drill hole **325b**), and a third portion or top portion (e.g., top drill hole **325c**). In some cases, as shown in FIG. 3B, the first portion or first drill hole **325a** has a first width or first diameter **W1**, the second portion or second drill hole **325b** has a second width or second diameter **W2**, and the third portion or third drill hole **325c** has a third width or third diameter **W3**. The via **330** is not limited to only three portions, three drill holes, and/or three widths and may have more or less than three portions, three drill holes, and/or three

widths.

[0061] In some cases, the first width (or bottom width) **W1** of the first drill hole **325a** and the third width (or top width) **W3** of the third drill hole **325c** might be a same width or approximately the same width. Alternatively, in other cases, the first width (or bottom width) **W1** of the first drill hole **325a** and the third width (or top width) **W3** of the third drill hole **325c** might be a different width. The second width (or middle width) **W2** of the second drill hole **325b** is different from the first width (or bottom width) **W1** of the first drill hole **325a** and the third width (or top width) **W3** of the third drill hole **325c**. The second width (or middle width) **W2** of the second drill hole **325b** is larger or has a greater width than the first width (or top width) **W1** of the first drill hole **325a** and the third width (or bottom width) **W3** of the third drill hole **325c**. In some cases, the second width (or middle width) **W2** of the second drill hole **325b** is formed using a larger drill than the drill used to form the first width (or bottom width) **W1** of the first drill hole **325a** and the third width (or top width) **W3** of the third drill hole **325c**.

[0062] By varying the width of the drill holes **325a**, **325b**, and **325c** and/or via **330** such that the via **330** has a larger width in a middle portion or middle drill hole **325b** of the via **330** and by having a multilayer core **315**, impedance variations seen by the electrical signal as it travels through the substrate **300** by the via **330** may be minimized. For example, by having a bigger diameter in the middle portion or middle drill hole **325b** of the via **330** and smaller diameter at the two ends (e.g., bottom portion or bottom drill hole **325a** and top portion or top drill hole **325c**) of the via **330** and a multilayer core **315**, this arrangement helps to minimize the inductive peak seen in the impedance variation of the substrate **300** and/or via **330**. This in turn may improve insertion and return loss of the substrate **300** and/or via **330**. Further, by implementing variable width via **330** in a multilayer core **315**, resonance seen in the insertion loss in the frequency bandwidth of interest may be minimized or eliminated. Additionally, by implementing a variable width via **330** in a multilayer core **315**, package warpage of the chip or semiconductor may be reduced.

[0063] Different arrangements of substrate **300**, multilayer core **315**, and vias **330** are illustrated below with respect to FIGS. 4-7.

[0064] FIGS. 4A and 4B (collectively, FIG. 4) are schematic cross-sectional views of a substrate **400a** and **400b** (collectively, substrate **400**) of a semiconductor or chip module, in accordance with various embodiments. The substrate **400** may be similar to the substrate **200** of FIGS. 2 and/or **300** of FIG. 3, except instead of only having one via shown, FIG. 4 has at least two vias **425a** and **425b**. In some cases, functionalities, features, modifications, alterations and/or the like described in FIGS. 1-3 may also be applied to FIG. 4 and vice versa. It should be noted that the various components of the substrate **400** are schematically illustrated in FIG. 4, and that modifications to the various components and other arrangements of the substrate **400** may be possible and in accordance with the various embodiments.

[0065] The substrate **400** may include a first layer **405** (similar to first layer **205** of FIG. 2, first layer **205** of FIG. 2, or first layer **305** of FIG. 3) and a second layer **410** (similar to second layer **110** of FIG. 1, second layer **210** of FIG. 2, or second layer **310** of FIG. 3). A core **415** is located between the first layer **405** and the second layer **410** of the substrate **400**. The core **415** may have one core layer **420** as shown in FIG. 4A and/or more than one layer (e.g., core layers **420a**, **420b**, and/or **420c**) as shown in FIG. 4B.

[0066] The core **415** may have two or more vias **425a** and **425b**. Each via **425a** and **425b** may extend along an axis (not shown) which is normal to a plane defined by a planar surface of the core **415**. In some cases, as shown in FIG. 4B, each layer **420** of the core **415** may have a corresponding first drill hole (e.g., drill holes **430a**, **430b**, and **430c**) forming first via **425a** and a corresponding second drill hole (e.g., drill holes **435a**, **435b**, and **435c**) forming second via **425b**. The vias **425a** and **425b** create an electrical connection from the first layer **405** to the second layer **410** of the substrate **400**. Although two vias **425a** and **425b** are shown in FIGS. 4A and 4B and six drill holes **430a**, **430b**, and **430c** and **435a**, **435b**, and **435c** are shown in FIG. 4B, there may be more or less

vias located in core **415** and/or drill holes located in each core layer **420**.

[0067] In some instances, the vias **425a** and **425b** might optionally include one or more via pads **440**. Via pads **440** are electrically coupled to first drill holes **430a-430c** or second drill holes **435a-435c** and/or to vias **425a** and **425b**. In some cases, the via pads **440** might electrically couple the vias **425a** and **425b** to one or more electrically conductive components (e.g., pins, pads, solder balls, etc.) located on the first layer **405** and the second layer **410**, to a conductive coating located on the first layer **405** and the second layer **410**, or to the first layer **405** and the second layer **410** formed from a conductive material, and/or the like. In this manner, an electrical connection from the first layer **405** to the second layer **410** of the substrate **400** may be created by vias **425a** and **425b**.

[0068] In some instances, the openings of first drill holes **430a**, **430b**, and **430c**, the openings of second drill holes **435a**, **435b**, and **435c**, the openings of vias **425a** and/or **425b**, and/or via pads **440** might be circle-shaped, oval-shaped, square-shaped, rectangular-shaped, and/or the like. The first via **425a** includes a first portion or bottom portion **445a** (e.g., bottom drill hole **430a**), a second portion or middle portion **445b** (e.g., middle drill hole **430b**), and a third portion or top portion **445c** (e.g., top drill hole **430c**). The second via **425b** includes a first portion or bottom portion **450a** (e.g., bottom drill hole **435a**), a second portion or middle portion **450b** (e.g., middle drill hole **435b**), and a third portion or top portion **450c** (e.g., top drill hole **435c**).

[0069] In some cases, the first portion **445a** or bottom drill hole **430a** of the first via **425a** and a first portion **450a** or bottom drill hole **435a** of the second vias **425b** has a first width or first diameter **W1**. The second portion **445b** or middle drill hole **430b** of the first via **425a** and the second portion **450b** or middle drill hole **435b** of the second via **425b** has a second width or second diameter **W2**. The third portion **445c** or top drill hole **430c** of the first via **425a** and the third portion **450c** or top drill hole **435c** of the second via **425b** has a third width or third diameter **W3**. The vias **425a** and **425b** are not limited to only three portions, three widths, and/or three drill holes and may have more or less than three portions, three widths, and/or three drill holes.

[0070] In various embodiments, first via **425a** and second via **425b** might be differential pair vias. The differential pair vias may be routed side-by-side through the core **415** and have the same or approximately the same dimensions. Each via **425a** and **425b** may carry a signal with a same magnitude and an opposite polarity. For example, first via **425a** may be operable to carry an electrical signal with a first polarity and second via **425b** may be operable to carry a second electrical signal with an opposite polarity from the first electrical signal. The differential pair vias help lower electromagnetic radiation and pair-to-pair crosstalk.

[0071] In some cases, the first width (or bottom width) **W1** and the third width (or top width) **W3** of each differential via **425a** and **425b** might be a same width or approximately the same width. Alternatively, in other cases, the first width (or bottom width) **W1** and the third width (or top width) **W3** of each vias **425a** and **425b** might be a different width. The second width (or middle width) **W2** of each differential via **425a** and **425b** is different from the first width (or bottom width) **W1** and the third width (or top width) **W3** of each via **425a** and **425b**. The second width (or middle width) **W2** of each differential via **425a** and **425b** is larger or has a greater width than the first width (or top width) **W1** and the third width (or bottom width) **W3** of each via **425a** and **425b**. By having a larger second width (or middle width) **W2**, the middle portions **445b** and **450b** and/or the middle drill holes **430b** and **435b** of vias **425a** and **425b** may be brought closer together and/or a distance between the middle portions **445b** and **450b** and/or the middle drill holes **430b** and **435b** of vias **425a** and **425b** may be reduced.

[0072] Further advantages may be realized by routing two differential pair vias with variable widths. For example, electromagnetic radiation and pair-to-pair crosstalk may further be minimized as the variable width of the middle portions **445b** and **450b** or middle drill holes **430b** and **435b** of the vias **425a** and **425b** brings the differential vias **425a** and **425b** closer together. Additionally, impedance variations seen by the electrical signals as they travel through the substrate **400** by the

differential vias **425a** and **425b** may be minimized. For example, by having a bigger diameter in middle portions **445b** and **450b** or middle drill holes **430b** and **435b** of the vias **425a** and **425b** and smaller diameter at the two ends (e.g., bottom portions **445a** and **450a** or bottom drill holes **430a** and **435a** and top portions **445c** and **450c** or top drill holes **430c** and **435c**) of the differential vias **425a** and **425b**, this arrangement helps to minimize the inductive peak seen in the impedance variation of the substrate **400**. This in turn may improve insertion and return loss of the substrate **400**. Further, by implementing variable width differential vias **425a** and **425b** in a multilayer core **415**, as shown in FIG. 4B, resonance seen in the insertion loss in the frequency bandwidth of interest may be minimized or eliminated.

[0073] FIGS. 5A and 5B (collectively, FIG. 5) are schematic cross-sectional views of a substrate **500a** and **500b** (collectively, substrate **500**) of a semiconductor or chip module, in accordance with various embodiments. The substrate **500** may be similar to substrate **100** of FIG. 1, substrate **200** of FIG. 2, substrate **300** of FIG. 3, and/or substrate **400** of FIG. 4, except instead of only having a constant width via and/or a variable width via, FIG. 5 has at least one offset via. In some cases, functionalities, features, modifications, alterations and/or the like described in FIGS. 1-4 may also be applied to FIG. 5 and vice versa. It should be noted that the various components of the substrate **500** are schematically illustrated in FIG. 5, and that modifications to the various components and other arrangements of the substrate **500** may be possible and in accordance with the various embodiments.

[0074] The substrate **500** may include a first layer **505** (similar to first layer **105** of FIG. 1, first layer **205** of FIG. 2, first layer **305** of FIG. 3, or first layer **405** of FIG. 4) and a second layer **510** (similar to second layer **110** of FIG. 1, second layer **210** of FIG. 2, second layer **310** of FIG. 3, or second layer **410** of FIG. 4). A core **515** is located between the first layer **505** and the second layer **510** of the substrate **500**. In FIG. 5, the core **515** is only one layer, but is not limited to only one layer.

[0075] The core **515** may include a via **520**. The via **520** may extend along an axis A-A (shown in FIG. 5A) which is normal to a plane defined by a planar surface of the core **515**. It should be noted that the core **515** is not limited to only one via **520** and may include more than one via **520**.

[0076] The via **520** may be formed in the core **515** by drilling a hole in the core **515** and plating or coating an inner surface of the hole with an electrically conductive material. An electrical signal is conducted from one electrically conductive component or layer **505** to another electrically conductive component or layer **510** of a substrate **500** by the via **520**.

[0077] In some instances, the opening or drill hole **525** of via **520** might be circle-shaped, oval-shaped, square-shaped, rectangular-shaped, and/or the like. Typically, the opening or drill hole **525** of the via **520** will be circle-shaped or oval-shaped. The via **520** includes a first portion or bottom portion **530**, a second portion or middle portion **535**, and a third portion or top portion **540**.

[0078] In some cases, a length of each of the first portion **530**, the second portion **535**, and the third portion **540** may be the same or approximately the same. Alternatively, a length of each of the first portion **530**, the second portion **535**, and the third portion **540** may vary. For example, the length of the middle portion **535** may be longer than the length of the bottom portion **530** and the top portion **540**. Alternatively, in some cases, the length of the middle portion **535** may be shorter than the length of the bottom portion **530** and the top portion **540**. Each of the lengths of the first portion **530**, the second portion **535**, and the third portion **540** may be adjusted to reduce the impedance variation of the substrate **500**. In some cases, when core **515** is a multilayer core, the length of the portions of via **520** might vary based on the thickness of each core layer.

[0079] In some cases, the first portion **530** has a first width or a first diameter **W1** (shown in FIG. 5B), the second portion **535** has a second width or a second diameter **W2** (shown in FIG. 5B), and the third portion **540** has a third width or a third diameter **W3** (shown in FIG. 5B). Although the widths are only shown in FIG. 5B, a person of ordinary skill in the art would know how to identify the widths in the FIG. 5A. The via **520** is not limited to only three portions and/or three widths and

may have more or less than three portions and/or three widths.

[0080] In some cases, the first width (or bottom width) **W1**, the second width (or middle width) **W2**, and the third width (or bottom width) **W3** might be a same width or approximately the same width. Alternatively, in other cases, the first width (or bottom width) **W1**, the second width (or middle width) **W2**, and the third width (or top width) **W3** might be different widths. In some instances, as shown in FIG. 5B, the second width (or middle width) **W2** is different from the first width (or bottom width) **W1** and the third width (or top width) **W3**. The second width (or middle width) **W2** may be larger or has a greater width than the first width (or bottom width) **W1** and the third width (or top width) **W3** or the second width (or middle width) **W2** may be smaller than the first width (or bottom width) **W1** and the third width (or top width) **W3**.

[0081] By varying the width of the via **520** such that the via **520** has a larger width in a middle portion of the via **520**, impedance variations seen by the electrical signal as it travels through the substrate **500** by the via **520** may be minimized. For example, by having a bigger diameter in the middle portion **535** of the via **520** and smaller diameter at the two ends (e.g., bottom portion **530** and top portion **540**) of the via **520**, this arrangement helps to minimize the inductive peak seen in the impedance variation profile of the substrate **500** and/or via **520**.

[0082] In some embodiments, a cross-section of the via **520** might vary along the length L_y of the via **520** as the via **520** extends through the core **515**. The cross-section of the via **520** may vary due to offsetting one or more portions of the via **520**. As shown in FIGS. 5A and 5B, the middle portion **535** of the via **520**, is offset from the bottom portion **530** and the top portion **540** of the via **520**. In some cases, the bottom portion **530** and the top portion **540** of the via **520** are aligned or approximately aligned along a vertical axis A-A extending through a middle of the bottom portion **530** and the top portion **540** of the via **520** while the middle portion **535** is offset from the bottom portion **530** and the top portion **540** of the via **520**. In various instances, as shown in FIG. 5A, a second edge **545b** of the middle portion **535** might align with a first edge **545a** of the bottom portion **530** and a third edge **545c** of the top portion **540** of the via **520**. In other cases, as shown in FIG. 5B, a second edge **545b** of the middle portion **535** might align with a vertical axis (not shown, but shown in FIG. 5A) extending through a middle of the bottom portion **530** and the top portion **540** of the via **520**. The amount that the middle portion **535** of the via **520** is offset from the bottom portion **530** and the top portion **540** of the via **520** might vary in order to minimize impedance variations seen by the electrical signal as it travels through the substrate **500** by the via **520**.

[0083] By offsetting the middle portion **535** of the via **520** from the bottom portion **530** of the via **520** and the top portion **540** of the via **520**, impedance variations seen by the electrical signal as it travels through the substrate **500** by the via **520** may be minimized. For example, by offsetting the middle portion **535** of the via **520** from the bottom portion **530** of the via **520** and the top portion **540** of the via **520**, this arrangement helps to minimize the inductive peak seen in the impedance variation profile of the substrate **500** and/or via **520**. This in turn may improve insertion and return loss of the substrate **500** and/or via **520**.

[0084] Different arrangements of substrate **500**, core **515**, and via **520** are illustrated below with respect to FIGS. 6 and 7.

[0085] FIGS. 6A and 6B (collectively, FIG. 6) are schematic cross-sectional views of a substrate **600a** and **600b** (collectively, substrate **600**) of a semiconductor or chip module, in accordance with various embodiments. The substrate **600** may be similar to substrate **100** of FIG. 1, substrate **200** of FIG. 2, substrate **300** of FIG. 3, substrate **400** of FIG. 4, and/or substrate **500** of FIG. 5, except instead of only having a constant width via and/or a variable width via, FIG. 6 has at least one offset via within a multilayer core. It should be noted that the various components of the substrate **600** are schematically illustrated in FIG. 6, and that modifications to the various components and other arrangements of the substrate **600** may be possible and in accordance with the various embodiments.

[0086] The substrate **600** may include a first layer **605** (similar to first layer **105** of FIG. 1, first

layer **205** of FIG. 2, first layer **305** of FIG. 3, first layer **405** of FIG. 4, or first layer **505** of FIG. 5) and a second layer **610** (similar to second layer **110** of FIG. 1, second layer **210** of FIG. 2, second layer **310** of FIG. 3, second layer **410** of FIG. 4, or second layer **510** of FIG. 5). A core **615** is located between the first layer **605** and the second layer **610** of the substrate **600**. The core **615** may have one or more layers **620a**, **620b**, and **620c** (collectively, core layers **620**). Although three core layers **620** are shown in FIG. 6, there may be more or less core layers **620** located in substrate **600**. [0087] Each core layer **620a**, **620b**, and **620c** may have a corresponding drill hole **625a**, **625b**, and **625c**. Although three drill holes **625a**, **625b**, and **625c** are shown in FIG. 6, there may be more or less drill holes located in each core layer **620a**, **620b**, and **620c**. The three drill holes **625a**, **625b**, and **625c** might electrically couple together to form a via **630** through the core layers **620a**, **620b**, and **620c** of the core **615**. The via **630** may extend along an axis A-A (shown in FIG. 6A) which is normal to a plane defined by a planar surface of the core **615**. It should be noted that the core **615** is not limited to only one via **630** and may include more than one via. Each core layer **620**, drill hole **625a-625c**, and via **630** may be formed in a manner similar to the manufacturing method recited in FIG. 8.

[0088] In some instances, the first drill hole **625a** might optionally include a first via pad **635a** and a second via pad **635b**, the second drill hole **625b** might optionally include the second via pad **635b** and a third via pad **635c**, and the third drill hole **625c** might optionally include the third via pad **635c** and a fourth via pad **635d**. Via pads **635a-635d** are electrically coupled to drill holes **625a-625c** and/or via **630**. In some cases, via pads **635a-635d** may form electrical connections or pathways between drill holes **625a-625c**. In some instances, vias pads **635a-635d** run along a first parallel axis arranged parallel to the plane defined by the planar surface on top of a corresponding core layer **620a-620c** or between two corresponding core layer **620a-620c** to electrically couple drill holes **625a-625c** together and form via **630**.

[0089] The via **630** includes a first portion or bottom portion (e.g., bottom drill hole **625a**), a second portion or middle portion (e.g., middle drill hole **625b**), and a third portion or top portion (e.g., top drill hole **625c**). In some cases, as shown in FIG. 6B, the first portion or first drill hole **625a** has a first width or first diameter **W1**, the second portion or second drill hole **625b** has a second width or second diameter **W2**, and the third portion or third drill hole **625c** has a third width or third diameter **W3**. The via **630** is not limited to only three portions, three drill holes, and/or three widths and may have more or less than three portions, three drill holes, and/or three widths. [0090] As shown in FIGS. 6A and 6B, the middle drill hole **625b** of the via **630**, is offset from the bottom drill hole **625a** and the top drill hole **625c** of the via **630**. In some cases, the bottom drill hole **625a** and the top drill hole **625c** of the via **630** are aligned or approximately aligned along a vertical axis A-A extending through a middle of the bottom drill hole **625a** and the top drill hole **625c** of the via **630** while the middle drill hole **625b** is offset from the bottom drill hole **625a** and the top drill hole **625c** of the via **630**. In some instances, as shown in FIG. 6B, a second edge **640b** of the middle drill hole **625b** might align with a first edge **640a** of the bottom drill hole **625a** and a third edge **640c** of the top drill hole **625c** and a width **W2** of the middle drill hole **625b** might cause the middle drill hole **625b** to be offset from the bottom drill hole **625a** and the top drill hole **625c**. The amount that the middle drill hole **625b** of the via **630** is offset from the bottom drill hole **625a** and the top drill hole **625c** of the via **630** might vary in order to minimize impedance variations seen by the electrical signal as it travels through the substrate **600** by the via **630**.

[0091] By offsetting the middle drill hole **625b** of the via **630** from the bottom drill hole **625a** of the via **630** and the top drill hole **625c** of the via **630**, impedance variations seen by the electrical signal as it travels through the substrate **600** by the via **630** may be minimized. For example, by offsetting the middle drill hole **625b** of the via **630** from the bottom drill hole **625a** of the via **630** and the top drill hole **625c** of the via **630**, this arrangement helps to minimize the inductive peak seen in the impedance variation of the substrate **600** and/or via **630**. This in turn may improve insertion and return loss of the substrate **600** and/or via **630**. Further, by implementing an offset via

630 in a multilayer core **615**, package warpage of the chip or semiconductor may be reduced and resonance may be decreased.

[0092] FIGS. 7A and 7B (collectively, FIG. 7) are schematic cross-sectional views of a substrate **700a** and **700b** (collectively, substrate **700**) of a semiconductor or chip module, in accordance with various embodiments. The substrate **700** may be similar to the substrate **500** of FIG. 5 and/or substrate **600** of FIG. 6, except instead of only having one via, FIG. 7 has at least two vias. In some cases, functionalities, features, modifications, alterations and/or the like described in FIGS. 1-6 may also be applied to FIG. 7 and vice versa. It should be noted that the various components of the substrate **700** are schematically illustrated in FIG. 7, and that modifications to the various components and other arrangements of the substrate **700** may be possible and in accordance with the various embodiments.

[0093] The substrate **700** may include a first layer **705** (similar to first layer **105** of FIG. 1, first layer **205** of FIG. 2, first layer **305** of FIG. 3, first layer **405** of FIG. 4, first layer **505** of FIG. 5, or first layer **605** of FIG. 6) and a second layer **710** (similar to second layer **110** of FIG. 1, second layer **210** of FIG. 2, second layer **310** of FIG. 3, second layer **410** of FIG. 4, second layer **510** of FIG. 5, or second layer **610** of FIG. 6). A core **715** is located between the first layer **705** and the second layer **710** of the substrate **700**. The core **715** may have one core layer **720** as shown in FIG. 7A and/or more than one layer (e.g., core layers **720a**, **720b**, and/or **720c**) as shown in FIG. 7B.

[0094] The core **715** may have two or more vias **725a** and **725b**. Each via **725a** and **725b** may extend along an axis (not shown) which is normal to a plane defined by a planar surface of the core **715**. In some cases, as shown in FIG. 7B, each layer **720** of the core **715** may have a corresponding first drill hole (e.g., drill holes **730a**, **730b**, and **730c**) forming first via **725a** and a corresponding second drill hole (e.g., drill holes **735a**, **735b**, and **735c**) forming second via **725b**. Although two vias **725a** and **725b** are shown in FIGS. 7A and 7B and six drill holes **730a**, **730b**, and **730c** and holes **735a**, **735b**, and **735c** are shown in FIG. 7B, there may be more or less vias located in core **715** and/or drill holes located in each core layer **720**. In some cases, as shown in FIG. 7B, the three first drill holes **730a**, **730b**, and **730c** might electrically couple together to form a first via **725a** through the core layers **720a**, **720b**, and/or **720c** and the three second drill holes **735a**, **735b**, and **735c** might electrically couple together to form a second via **725b** through the core layers **720a**, **720b**, and/or **720c**.

[0095] In some instances, the vias **725a** and **725b** might optionally include one or more via pads **740**. Via pads **740** are electrically coupled to first drill holes **730a-730c** or second drill holes **735a-735c** and/or to vias **725a** and **725b**. In some cases, the via pads **740** might electrically couple the vias **725a** and **725b** to one or more electrically conductive components (e.g., pins, pads, solder balls, etc.) located on the first layer **705** and the second layer **710**, to a conductive coating located on the first layer **705** and the second layer **710**, or to the first layer **705** and the second layer **710** formed from a conductive material, and/or the like. In this manner, an electrical connection from the first layer **705** to the second layer **710** of the substrate **700** may be created by vias **725a** and **725b**.

[0096] The first via **725a** includes a first portion or bottom portion **745a** (e.g., bottom drill hole **730a**), a second portion or middle portion **745b** (e.g., middle drill hole **730b**), and a third portion or top portion **745c** (e.g., top drill hole **730c**). The second via **725b** includes a first portion or bottom portion **750a** (e.g., bottom drill hole **735a**), a second portion or middle portion **750b** (e.g., middle drill hole **735b**), and a third portion or top portion **750c** (e.g., top drill hole **735c**).

[0097] In some cases, first via **725a** and second via **725b** might be a pair of vias (e.g., a pair of differential pair vias). The differential pair vias may be routed side-by-side through the core **715** and have the same or approximately the same dimensions. Each via **725a** and second via **725b** may carry a signal with a same magnitude and an opposite polarity from the other via. For example, via **725a** may be operable to carry an electrical signal with a first polarity and via **725b** may be operable to carry a second electrical signal with an opposite polarity from the first electrical signal.

The differential pair vias help lower electromagnetic radiation and pair-to-pair crosstalk.

[0098] In some embodiments, as shown in FIGS. 7A and 7B, the middle portion **745b** or middle drill hole **730b** of the first via **725a** is offset from a bottom portion **745a** or bottom drill hole **730a** and the top portion **740c** of the via **725a**. Additionally, the middle portion **750b** or middle drill hole **735b** of the second via **725b** is offset from a bottom portion **750a** or bottom drill hole **735a** and the top portion **750c** or top drill hole **735c** of the via **725b**. The middle portion **745b** or middle drill hole **730b** of the first via **725a** is offset toward the middle portion **750b** or middle drill hole **735b** of the second via **725b**. The middle portion **750b** or middle drill hole **735b** and of the second via **725b** is offset toward the middle portion **745b** or middle drill hole **730b** of the first via **725a**.

[0099] In some cases, as shown in FIG. 7B, the middle drill hole **730b** of the first via **725a** is completely offset (e.g., does not overlap) from the bottom drill hole **730a** and the top drill hole **730c**. The via pads **740** run along a first parallel axis parallel to the plane defined by the planar surface of each of the two or more core layers to electrically couple the middle drill hole **730b** of the first via **725a** to the bottom drill hole **730a** and the top drill hole **730c**. Additionally, the middle drill hole **735b** of the second via **725b** is completely offset (e.g., does not overlap) from the bottom drill hole **735a** and the top drill hole **735c**. The via pads **740** run along a first parallel axis parallel to the plane defined by the planar surface of each of the two or more core layers to electrically couple the middle drill hole **735b** of the first via **725a** to the bottom drill hole **730a** and the top drill hole **730c**.

[0100] By offsetting the middle portions **745b** and **750b** and/or the middle drill holes **730b** and **735b** of vias **725a** and **725b** toward each other, the middle portions **745b** and **750b** and/or the middle drill holes **730b** and **735b** of vias **725a** and **725b** may be brought closer together and/or a distance between the middle portions **745b** and **750b** and/or the middle drill holes **730b** and **735b** of vias **725a** and **725b** may be reduced. The amount that the middle portions **745b** and **750b** and/or the middle drill holes **730b** and **735b** of vias **725a** and **725b** is offset from the bottom portion or bottom drill hole and/or the top portion or top drill hole of vias **725a** and **725b** might vary in order to minimize impedance variations seen by the electrical signal as it travels through the substrate **700** by the vias **725a** and **725b**.

[0101] Additional advantages may be realized by routing two differential pair vias with offset middle portions or offset middle drill holes. For example, electromagnetic radiation and pair-to-pair crosstalk may further be minimized as the offset middle portion or offset middle drill hole of the vias **725a** and **725b** brings the differential vias **725a** and **725b** closer together. Additionally, impedance variations seen by the electrical signals as they travel through the substrate **700** by the vias **725a** and **725b** may be minimized. For example, by offsetting the middle portions **745b** and **750c** of the vias **725a** and **725b** from the two ends (e.g., bottom portions **745a** and **750a** and top portions **745c** and **750c**) of the vias **725a** and **725b**, this arrangement helps to minimize the inductive peak seen in the impedance variation of the substrate **700**. This in turn may improve insertion and return loss of the substrate **700**. Further, by implementing offset vias **725a** and **725b** in a multilayer core **715**, as shown in FIG. 7B, resonance seen in the insertion loss in the frequency bandwidth of interest may be minimized or eliminated.

[0102] FIG. 8 is a flow diagram of a method **800** of manufacturing a multilayer core, a variable diameter via, and/or an offset via, in accordance with various embodiments. In some instances, the method **800** might be used to manufacture a multilayer core, a variable diameter via, and/or an offset via, in accordance with FIGS. 1-7. The method **800** may begin, at block **805**, by forming at least one first core layer of a core. The core might be a core of a substrate. The substrate might be a printed circuit board ("PCB"). Forming the first core layer of the core may include, without limitation, various additive and/or subtractive manufacturing processes, as known to those skilled in the art. In further examples, manufacturing techniques that are being developed and/or yet to be developed may be utilized to form the first core layer. Thus, it is to be understood that the method **800** is not limited to any particular method of manufacturing a particular core.

[0103] The method **800** continues, at block **810**, by drilling at least one first drill hole into the first core layer of the core. The at least one first drill might have a first width. The openings of the drill hole might be circle-shaped, oval-shaped, square-shaped, rectangular-shaped, and/or the like.

[0104] The method **800** continues, at block **815**, by forming at least one other core layer. The at least one other core layer is layered or formed on top of or below the at least one first core layer along an axis normal to a plane defined by a planar surface of the at least one first core layer. Like the first core layer, formation of the other core layer is not limited to any particular manufacturing technique.

[0105] At block **820**, the method **800** may continue by drilling at least one other drill hole into the other core layer of the core. The at least one other drill hole might have a second width which may be different from or the same as the first width. The openings of drill hole might be circle-shaped, oval-shaped, square-shaped, rectangular-shaped, and/or the like.

[0106] The method **800** continues, at block **825**, by electrically coupling the at least one first drill hole with the at least one other drill hole to create a via through the core. In some cases, the via has a cross-section that varies along a length of the via as the via extends through the two or more core layers. In some embodiments, the cross-section may vary based on having a width of the at least one first drill hole different from a width of the at least one other drill hole creating a via with a variable width in accordance with FIGS. 2-4. Additionally or alternatively, the cross-section may vary based on offsetting the at least one first drill hole from the at least one other drill hole creating an offset via in accordance with FIGS. 5-7.

[0107] The techniques and processes described above with respect to various embodiments may be used to manufacture the substrates **100**, **200**, **200**, **300**, **400**, **500**, **600**, and **700**, and/or components thereof, as described herein.

[0108] While some features and aspects have been described with respect to the embodiments, one skilled in the art will recognize that numerous modifications are possible. For example, the methods and processes described herein may be implemented using hardware components, custom integrated circuits (ICs), programmable logic, and/or any combination thereof. Further, while various methods and processes described herein may be described with respect to particular structural and/or functional components for ease of description, methods provided by various embodiments are not limited to any particular structural and/or functional architecture but instead can be implemented in any suitable hardware configuration. Similarly, while some functionality is ascribed to one or more system components, unless the context dictates otherwise, this functionality can be distributed among various other system components in accordance with the several embodiments.

[0109] Moreover, while the procedures of the methods and processes described herein are described in a particular order for ease of description, unless the context dictates otherwise, various procedures may be reordered, added, and/or omitted in accordance with various embodiments. Moreover, the procedures described with respect to one method or process may be incorporated within other described methods or processes; likewise, system components described according to a particular structural architecture and/or with respect to one system may be organized in alternative structural architectures and/or incorporated within other described systems. Hence, while various embodiments are described with or without some features for ease of description and to illustrate aspects of those embodiments, the various components and/or features described herein with respect to a particular embodiment can be substituted, added and/or subtracted from among other described embodiments, unless the context dictates otherwise. Consequently, although several embodiments are described above, it will be appreciated that the invention is intended to cover all modifications and equivalents within the scope of the following claims.

Claims

1. An apparatus comprising: a substrate, wherein the substrate comprises: a core comprising: a first via extending through the core, wherein the first via includes a first portion offset from a second portion of the first via.
2. The apparatus of claim 1, wherein the first via extends through the core along an axis normal to a plane defined by a planar surface of the core.
3. The apparatus of claim 1, wherein the first via includes a third portion offset from the second portion of the first via, wherein the second portion is between the first portion and the third portion.
4. The apparatus of claim 3, wherein the first portion and the third portion are aligned along a vertical axis extending through a middle portion of the first portion and the third portion.
5. The apparatus of claim 3, wherein a second width of the second portion is greater than a first width of the first portion and a third width of the third portion.
6. The apparatus of claim 5, wherein the first width of the first portion is a same width as the third width of the third portion.
7. The apparatus of claim 3, wherein a second edge of the second portion aligns with a first edge of first portion and a third edge of the third portion.
8. The apparatus of claim 3, wherein a second edge of the second portion aligns with a first middle of the first portion and a second middle of the third portion.
9. The apparatus of claim 3, wherein a second edge of the second portion is spaced apart from and does not overlap a first edge of first portion and a third edge of the third portion.
10. The apparatus of claim 9, wherein the first portion is coupled to the second portion using a first via pad and wherein the second portion is coupled to the third portion using a second via pad.
11. The apparatus of claim 3, wherein the first portion, the second portion, and the third portion are a same length.
12. The apparatus of claim 3, wherein a second length of the second portion is greater than a first length of the first portion and a third length of the third portion.
13. The apparatus of claim 1, wherein the first via is part of a pair of vias, wherein a first portion of a second via of the pair of vias is offset from a second portion of the second via.
14. The apparatus of claim 13, wherein the second portion of the first via is offset toward the second portion of the second via, and wherein the second portion of the second via is offset toward the second portion of the first via.
15. The apparatus of claim 14, wherein the second portion of the first via is a first middle portion of the first via and the second portion of the second via is a second middle portion of the second via, and wherein the first middle portion of the first via is offset toward the second middle portion of the second via and the second middle portion of the second via is offset toward the first middle portion of the first via.
16. The apparatus of claim 15, wherein the first middle portion of the first via is a same length as the second middle portion of the second via.
17. The apparatus of claim 15, wherein the first via is operable to carry a first electrical signal with a first polarity and the second via is operable to carry a second electrical signal with an opposite polarity from the first electrical signal.
18. A semiconductor device comprising: a substrate, wherein the substrate comprises: two or more core layers, each core layer stacked on top of or below another core layer of the two or more core layers, each core layer comprising: at least one drill hole, wherein at least one first drill hole of at least one first core layer electrically connects with at least one other drill hole of at least one other core layer to form a first via through the two or more core layers, and wherein the first via includes a first portion of the first via that is offset from a second portion of the first via.
19. The semiconductor device of claim 18, wherein the second portion has a greater width than the first portion.
20. A method comprising: forming at least one first core layer of a core comprising at least one first

drill hole; forming at least one other core layer of the core comprising at least one other drill hole, wherein the at least one other core layer is layered on top of or below the at least one first core layer along an axis normal to a plane defined by a planar surface of the at least one first core layer; and electrically coupling the at least one first drill hole with the at least one other drill hole to create a via through the core, wherein the via includes a first portion of the via that is offset from a second portion of the via.
